

SE Lab 4 – Vibecoding

Name: Selvaganesh Arunmozhi

SRN: PES2UG24CS451

Class & Section: 5 H

Tool Used: Claude Code

Prompt

The screenshot shows the Claude Code interface with a purple header bar. The top bar has 'CLAUDE CODE' and 'CHAT' tabs, and icons for full screen, close, history, and a plus sign. The main area has a title '2048 game SE Lab 4' and a 'Show less' button. The prompt is in a dark box with a code icon and the filename 'README.md'. The response is a list of thoughts and actions, with some code blocks shown in a dark terminal window.

CLAUDE CODE CHAT

2048 game SE Lab 4

</> README.md

I am a game developer, and I need you to behave like my co-developer and tester on this project. I would need you to go through the README file, and implement the tasks accordingly. Notify once every task and test is done, and incrementally commit the changes to the main branch on the repository.

Do not make assumptions and proceed in case there is no clarity. Ask questions in case you need any requirements from my end.

Show less

- Thought for 0s
- I'll start by reading the README and getting a picture of the project.
- Bash** Read README and list project files and git state

```
IN cd "c:/Users/vk360/Desktop/College/Lab Work/SE/Lab 4 - Sept 30th/2048_SE" && cat 14_2048/README.md &
```

```
OUT Exit code 128
# Scenario 14 - 2048
```

- Thought for 1s
- The actual repo is the nested `14_2048/` folder (on `main`, with a GitHub remote). Let me inspect it and the code.
- Bash** Show inner repo state and source files

```
IN cd "c:/Users/vk360/Desktop/College/Lab Work/SE/Lab 4 - Sept 30th/2048_SE/14_2048" && git status && g
```

```
OUT On branch main
Your branch is up to date with 'origin/main'.
```

- Thought for 16s
- Before touching anything, I'll reproduce the defect the README describes by tracing `slide_line` on a few rows.

- **Bash** Reproduce merge bug and check input handling

```
IN cd "c:/Users/vk360/Desktop/College/Lab Work/SE/Lab 4 - Sept 30th/2048_SE/14_2048" && python -c "
from board import Board
for row in ([2,2,2,2],[2,2,4,0],[4,4,8,16],[2,0,2,0],[2,2,4,4],[8,8,8,0]):
```

```
OUT Exit code 1
[2, 2, 2, 2] -> [4, 4, 0, 0]
[2, 2, 4, 0] -> [8, 0, 0, 0]
```

- Thought for 11s
- I've confirmed the merge-chaining bug in `slide_line`, found that score is never tracked, and spotted an input validation flaw where `key not in "wasd"` incorrectly treats it as a substring check. Before writing fixes, I need a few decisions from you.

● Questions

Answered · 4 questions ^

Scoring is currently never updated. Should I use standard 2048 scoring (each merge adds the new tile's value to the score, e.g. $2+2 \rightarrow +4$)?

- ☒ **Standard scoring (Recommended)**
Each merge adds the merged tile's value to the score.
- ☐ **Different rule**
You'll specify a different scoring rule.

When the player undoes a move, what should happen to the Best score?

- ☐ **Keep peak (Recommended)**
Best is the highest score reached this run and is never reduced by undo.
- ☒ **Roll back too**
Undo also restores Best to its value before the undone move.

For the incremental commits to main: should I also push each commit to GitHub (origin/main at thisisselva18/14_2048)?

- ☐ **Commit + push each**
Commit locally and push to origin/main after each task.
- ☒ **Local commits only**
Commit to local main only; you push when ready.

Where should the automated tests live?

- ☐ **test_game.py in repo (Recommended)**
Add a stdlib unittest file (no dependencies) to the repo so the tests are committed and can be re-run.
- ☒ **Outside the repo**
I run tests from a scratch location; only game code is committed.

- Thought for 33s
- Noted the constraints and default assumptions for ambiguous README cases; starting Task 1 now, fixing the merge algorithm.

- **Edit** c:\Users\vk360\Desktop\College\Lab Work\SE\Lab 4 - Sept 30th\2048_SE\14_2048\board.py

Added 8 lines

```
def slide_line(line):
    """Compress a line towards index 0, merging equal neighbours.

    Each original tile takes part in at most one merge, so a tile
    created by a merge cannot merge again in the same move.
    """
    values = [x for x in line if x]
    result = []
    for value in values:
        if result and result[-1] == value:
            result[-1] *= 2 # intentional double-merge bug
        else:
            result.append(value)
    i = 0
```

- **Write** C:\Users\vk360\AppData\Local\Temp\claude\c--Users-vk360-Desktop-College-Lab-Work-SE-Lab-4---Sept-30th-2048-SE\12c5a845-7100-41b2-ac28-781bab0f064a\scratchpad\test_2048.py

83 lines

```
"""Tests for the 2048 lab project (kept outside the repo by request).
```

```
Run: python test_2048.py (set GAME_DIR to the 14_2048 folder if it moves)
```

- **Bash** Run Task 1 tests and check git status

```
IN cd "C:/Users/vk360/AppData/Local/Temp/claude/c--Users-vk360-Desktop-College-Lab-Work-SE-Lab-4---Sept-30th-2048-SE\12c5a845-7100-41b2-ac28-781bab0f064a\scratchpad"

OUT test_directional_moves (__main__.TestMergeSemantics.test_directional_moves) ... ok
test_four_equal (__main__.TestMergeSemantics.test_four_equal) ... ok
test_no_chain_merge_of_new_tile (__main__.TestMergeSemantics.test_no_chain_merge_of_new_tile) ... ok
```

- Thought for 0s
- All 7 Task 1 tests pass. Committing Task 1.

- **Bash** Commit Task 1 fix

```
IN cd "c:/Users/vk360/Desktop/College/Lab Work/SE/Lab 4 - Sept 30th/2048_SE/14_2048" && git add board.p
Fix chain-merge bug in slide_line (Task 1)

92460d7 Fix chain-merge bug in slide_line (Task 1)
```

- Thought for 1s
- **Task 1 is done and committed (92460d7).** Now **Task 2**: game-state handling. This covers scoring, win detection, no-move detection, tile-creation semantics, and exact command validation.

- **Bash** Add scoring and win check to board.py

```
IN cd "c:/Users/vk360/Desktop/College/Lab Work/SE/Lab 4 - Sept 30th/2048_SE/14_2048" && python - <<'EOF
import re
p = "board.py"

OUT warning: in the working copy of 'board.py', LF will be replaced by CRLF the next time Git touches it
board.py | 24 ++++++-----
1 file changed, 19 insertions(+), 5 deletions(-)
```

- Thought for 6s
- Board now tracks score and exposes `has_won()`. Next is the game loop: a testable `status()`, exact command validation, and a clean quit.

- **Bash** Add status(), input validation and clean quit to game.py

```
IN cd "c:/Users/vk360/Desktop/College/Lab Work/SE/Lab 4 - Sept 30th/2048_SE/14_2048" && python - <<'EOF
p = "game.py"
s = open(p, encoding="utf-8").read()

OUT warning: in the working copy of 'game.py', LF will be replaced by CRLF the next time Git touches it
diff --git a/game.py b/game.py
index 0c89a3c..08fb402 100644
```

- Thought for 1s
- The heredoc escaping turned `\n` into a literal line break inside the string, so I'll fix that line.
- **Read** c:\Users\vk360\Desktop\College\Lab Work\SE\Lab 4 - Sept 30th\2048_SE\14_2048\game.py (lines 47-52)

- **Edit** c:\Users\vk360\Desktop\College\Lab Work\SE\Lab 4 - Sept 30th\2048_SE\14_2048\game.py

Removed 1 line

```
[ ] print("
[ ] Goodbye.")
[ ] print("\nGoodbye.")
```

- Now I'll add the Task 2 tests: scoring, win, no-move boards, tile creation, invalid commands, and quitting. The interactive paths are driven through a real subprocess.
- **Edit** C:\Users\vk360\AppData\Local\Temp\claude\c--Users-vk360-Desktop-College-Lab-Work-SE-Lab-4---Sept-30th-2048-SE\12c5a845-7100-41b2-ac28-781bab0f064a\scratchpad\test_2048.py
Added 143 lines

```
[ ] def tiles(grid):
[ ]     return sum(1 for row in grid for x in row if x)
[ ]
[ ] def make_game(grid, score=0):
[ ]     from game import Game
[ ]     g = Game()
[ ]     g.board = make_board(grid, score)
[ ]     g.best_score = score
[ ]     return g
[ ]
[ ]
```

- **Edit** C:\Users\vk360\AppData\Local\Temp\claude\c--Users-vk360-Desktop-College-Lab-Work-SE-Lab-4---Sept-30th-2048-SE\12c5a845-7100-41b2-ac28-781bab0f064a\scratchpad\test_2048.py
Modified

```
[ ] grid[3][3] = 4 # equal to its left neighbour 4? grid[3][2] is 4
[ ] grid[3][3] = 4 # now equal to its left neighbour grid[3][2] == 4
```

- **Bash** Run all tests through Task 2

```
IN cd "C:/Users/vk360/AppData/Local/Temp/claude/c--Users-vk360-Desktop-College-Lab-Work-SE-Lab-4---Sept

OUT -----
Ran 22 tests in 0.386s
```

- Thought for 0s
- All 22 tests pass (7 from Task 1, 15 new). Committing Task 2.

- **Bash** Commit Task 2 changes

```
IN cd "C:/Users/vk360/Desktop/College/Lab Work/SE/Lab 4 - Sept 30th/2048_SE/14_2048" && git add board.p
Finish game-state handling (Task 2)

OUT warning: in the working copy of 'board.py', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'game.py', LF will be replaced by CRLF the next time Git touches it
M board.py
..
```

- Thought for 3s
 - ☒ **Task 2 is done and committed** (927480d).
- Now **Task 3**: one-level undo plus best score. Per your answer, the undo snapshot captures board, score **and** best, so undo rolls Best back too.

- **Read** c:\Users\vk360\Desktop\College\Lab Work\SE\Lab 4 - Sept 30th\2048_SE\14_2048\game.py

- **Edit** c:\Users\vk360\Desktop\College\Lab Work\SE\Lab 4 - Sept 30th\2048_SE\14_2048\game.py
Modified

```
self.best_score = 0
self.history = []
self.history = [] # at most one snapshot: one-level undo
```

- **Edit** c:\Users\vk360\Desktop\College\Lab Work\SE\Lab 4 - Sept 30th\2048_SE\14_2048\game.py
Added 16 lines

```
if key not in moves:
    return False
before = self.snapshot()
changed = moves[key]()
if changed: # an unchanged board never receives a new tile
    self.board.add_random_tile()
self.best_score = max(self.best_score, self.board.score)
self.history = [before]
return changed

def snapshot(self):
    return [row[:] for row in self.board.grid], self.board.score, self.best_score
```

- **Edit** c:\Users\vk360\Desktop\College\Lab Work\SE\Lab 4 - Sept 30th\2048_SE\14_2048\game.py
Removed 1 line

```
if key == "u":
    print("Undo is not implemented yet.")
    print("Undid last move." if self.undo() else "Nothing to undo.")
    continue
if key not in ("w", "a", "s", "d"):
    print(f"Unknown command {key!r}. Use W/A/S/D to move, U to undo, Q to quit.")
    continue
if self.move(key):
    self.best_score = max(self.best_score, self.board.score)
self.move(key)
```

- Now the Task 3 tests.

- **Edit** C:\Users\vk360\AppData\Local\Temp\claude\c--Users-vk360-Desktop-College-Lab-Work-SE-Lab-4---Sept-30th-2048-SE\12c5a845-7100-41b2-ac28-781bab0f064a\scratchpad\test_2048.py

Added 78 lines

```
class TestUndoAndBest(unittest.TestCase):
    """Task 3."""

    START = [[2, 2, 0, 0], [4, 0, 4, 0], [0] * 4, [0] * 4]

    def test_nothing_to_undo_at_start(self):
        g = make_game(self.START)
        self.assertFalse(g.undo())
        self.assertEqual(g.board.grid, self.START)

    def test_undo_restores_board_and_score_without_new_tile(self):
        g = make_game(self.START, score=20)
```

- **Bash** Run all tests through Task 3

```
IN cd "C:/Users/vk360/AppData/Local/Temp/claude/c--Users-vk360-Desktop-College-Lab-Work-SE-Lab-4---Sept-
```

```
OUT -----
Ran 31 tests in 0.473s
```

- Thought for 1s
- 31/31 pass. One test contains a redundant line that makes it confusing, so I'll remove it before committing.

- **Edit** C:\Users\vk360\AppData\Local\Temp\claude\c--Users-vk360-Desktop-College-Lab-Work-SE-Lab-4---Sept-30th-2048-SE\12c5a845-7100-41b2-ac28-781bab0f064a\scratchpad\test_2048.py

Removed 1 line

```
g.board.grid = [[4, 8, 16, 32], [0] * 4, [0] * 4, [0] * 4] # force a blocked row
g.history[0][0][0][:] = [2, 2, 0, 0] # snapshot still intact
self.assertFalse(g.move("a"))
```

- **Bash** Re-run tests and commit Task 3

```
IN cd "C:/Users/vk360/AppData/Local/Temp/claude/c--Users-vk360-Desktop-College-Lab-Work-SE-Lab-4---Sept-
Add one-level undo and best score tracking (Task 3)
```

```
OUT Ran 31 tests in 0.484s
```

```
OK
```

- Thought for 14s
- Task 3 is committed. Moving to Task 4: I'll have `Game.move` return move details (direction, points, merges) or `None`, then add `Game.play(key)` to convert that into a single feedback line, keeping `Board` itself silent.

- **Edit** c:\Users\vk360\Desktop\College\Lab Work\SE\Lab 4 - Sept 30th\2048_SE\14_2048\game.py

Added 7 lines

```
from collections import namedtuple
from board import Board
DIRECTIONS = {"w": "up", "a": "left", "s": "down", "d": "right"}
# Outcome of one accepted move; Game.move returns None when nothing changed.
MoveResult = namedtuple("MoveResult", "direction points merges")
```

- **Edit** c:\Users\vk360\Desktop\College\Lab Work\SE\Lab 4 - Sept 30th\2048_SE\14_2048\game.py

Added 18 lines

```
def move(self, key):
    """Apply one move. Returns a MoveResult, or None if the board did not change."""
    moves = {"a": self.board.move_left, "d": self.board.move_right,
            "w": self.board.move_up, "s": self.board.move_down}
    if key not in moves:
        return False
    return None
    before = self.snapshot()
    changed = moves[key]()
    if changed: # an unchanged board never receives a new tile
        tiles_before = self.tile_count()
    if not moves[key](): # an unchanged board never receives a new tile
```

- **Edit** c:\Users\vk360\Desktop\College\Lab Work\SE\Lab 4 - Sept 30th\2048_SE\14_2048\game.py

Modified

```
if key not in ("w", "a", "s", "d"):
    if key not in DIRECTIONS:
        print(f"Unknown command {key!r}. Use W/A/S/D to move, U to undo, Q to quit.")
        continue
    self.move(key)
    print(self.play(key))
```

- Thought for 3s

- Adding the Task 4 tests:

- **Edit** C:\Users\vk360\AppData\Local\Temp\claude\c--Users-vk360-Desktop-College-Lab-Work-SE-Lab-4---Sept-30th-2048-SE\12c5a845-7100-41b2-ac28-781bab0f064a\scratchpad\test_2048.py

Added 59 lines

```
class TestMoveFeedback(unittest.TestCase):
    """Task 4."""
    def test_move_result_values(self):
        g = make_game([[2, 2, 2, 2], [4, 0, 4, 0], [8, 0, 0, 0], [0] * 4])
        r = g.move("a")
        self.assertEqual(tuple(r), ("left", 4 + 4 + 8, 3))
    def test_plain_slide_feedback(self):
        g = make_game([[2, 0, 0, 0], [0] * 4, [0] * 4, [0] * 4])
        self.assertEqual(g.play("d"), "Moved right.")
```

- **Bash** Run all tests through Task 4

```
IN cd "C:/Users/vk360/AppData/Local/Temp/claude/c--Users-vk360-Desktop-College-Lab-Work-SE-Lab-4---Sept-30th-2048-SE\12c5a845-7100-41b2-ac28-781bab0f064a\scratchpad\test_2048.py"

OUT -----
Ran 38 tests in 0.707s
```

- Thought for 1s
- 38/38 pass. I'll clean up a stale test comment, then run a scripted play session to check the actual terminal output by eye.

- **Edit** c:\Users\vk360\AppData\Local\Temp\claude\c--Users-vk360-Desktop-College-Lab-Work-SE-Lab-4---Sept-30th-2048-SE\12c5a845-7100-41b2-ac28-781bab0f064a\scratchpad\test_2048.py

Removed 1 line

```
setup = "g.board.grid = [[2,2,0,0],[0]*4,[0]*4,[0]*4"
# a: merge; a: row now [4,x..] may still slide, so use a blocked direction
out, rc
```

- **Bash** Scripted play session to inspect output

```
IN cd "c:/Users/vk360/Desktop/College/Lab Work/SE/Lab 4 - Sept 30th/2048_SE/14_2048" && printf 'a\n\nf'
import random; random.seed(3)
from game import Game

OUT 2048 - W/A/S/D to move, U to undo, Q to quit.
```

- Thought for 3s
- The session checks out—the Task 1 fix works correctly in play ([2,2,4,_] merges to [4,4], not [8]), scoring and undo behave as expected, and a second undo is properly blocked. Committing Task 4 now.

- **Bash** Commit Task 4 and show history

```
IN cd "c:/Users/vk360/Desktop/College/Lab Work/SE/Lab 4 - Sept 30th/2048_SE/14_2048" && git add game.py
Add action-level move feedback (Task 4)

OUT warning: in the working copy of 'game.py', LF will be replaced by CRLF the next time Git touches it
30aa33e Add action-level move feedback (Task 4)
8a7a027 Add one-level undo and best score tracking (Task 3)
```

- Thought for 7s
- All four README tasks are done, and each has its own commit on local `main`. Nothing is pushed yet, as you asked. The 38 tests all pass, and I ran a scripted play session to check the terminal output.

Task	Commit	What changed
1. Merge semantics	<code>92460d7</code>	A tile can now merge only once per move. Before the fix I confirmed the bug: <code>[2,2,4,0]</code> gave <code>[8]</code> , <code>[2,2,4,4]</code> gave <code>[8,4]</code> and <code>[4,4,8,16]</code> gave <code>[32]</code> . Now they give <code>[4,4]</code> , <code>[4,8]</code> and <code>[8,8,16]</code> .
2. Game state	<code>927480d</code>	Merges now add the merged tile's value to the score; before, the score never went up. The game ends when a tile reaches 2048 or no moves are left, and a win beats a stuck board. Unchanged moves don't add a tile.
3. Undo + Best	<code>8a7a027</code>	One-level undo restores the board, score and Best, and removes the tile the move added. A second <code>u</code> says "Nothing to undo." Unchanged or invalid moves don't use up the undo.
4. Feedback	<code>30aa33e</code>	Each accepted move prints one line, e.g. <code>Moved left: 2 merges, +8 points.</code> A move that changes nothing prints <code>Can't move left: ...</code> and never says "Moved". Checking whether moves are possible prints nothing.

Two things I fixed along the way, both in the Task 2 commit:

- **Input check:** the old code checked whether the input appeared anywhere in the text "wasd", so pressing Enter on an empty line or typing `wa` was silently accepted and ignored. Both now get an "Unknown command" message.
- **Quitting:** `q`, Ctrl+Z and Ctrl+C now exit cleanly with a message instead of crashing.

Tests: they're in `test_2048.py` in my scratchpad, outside the repo as you asked

(`C:\Users\vk360\AppData\Local\Temp\claude\...\scratchpad\test_2048.py`). They use only the Python standard library and cover everything in the README's "Required testing" list. The invalid-command, quit, win, no-moves and undo checks run the real game loop with scripted input. Run them with `python test_2048.py`. Copy the file somewhere permanent if you want to keep it, since the scratchpad is temporary.

There are no new dependencies and nothing is saved to disk; all game state stays in memory.

The outer `2048_SE` folder is a separate Git repository on `master` with no commits. I didn't touch it; all commits are in the `14_2048` repository.

Still yours to do before submitting: push (`git push origin main`), record the after-changes video, and add the chat-history link. For the video, a good row to set up is `[2,2,4,_]` pressed left: it now gives `[4,4]`. Then press `u` to show undo.



ctrl esc to focus or unfocus Claude



🕒 44m

Opus 5.5 High

📄 README.md ✕



Auto



Ln 100, Col 111

Spaces: 4

UTF-8

CRLF

{ } Markdown



(⏮) Go Live

